

GATE QUESTION Q36 - DON'T CARE IMPLEMENTATION

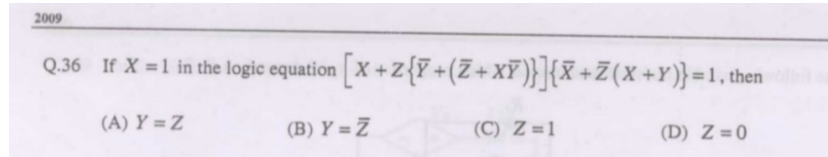


Figure 1: GATE Question Q36

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1 Components

Breadboard

Arduino Uno

IC 7474

IC 7447

Seven Segment Display

Jumper Wires

2 Boolean Simplification

Using don't care conditions, the minimized expression is

$$A = \bar{W}$$

The remaining outputs are

$$B = 0, \quad C = 0, \quad D = 0$$

When $W = 0$,

$$A = 1$$

Hence the seven segment display shows 1.

3 Hardware Connections

Arduino Pin D6 is connected to W.

Arduino Pin D7 is connected to X.

Arduino Pin D8 is connected to Y.

Arduino Pin D9 is connected to Z.

Arduino Pin D2 is connected to A of IC 7447.

Arduino Pin D3 is connected to B of IC 7447.

Arduino Pin D4 is connected to C of IC 7447.

Arduino Pin D5 is connected to D of IC 7447.

Arduino Pin D13 is connected to Clock.

4 Software Implementation

```
#include <Arduino.h>

int W,X,Y,Z;
int D,C,B,A;

void disp_dontcare(int D,int C,int B,int A)
{
    digitalWrite(2,A);
    digitalWrite(3,B);
    digitalWrite(4,C);
    digitalWrite(5,D);
}

void setup()
{
    pinMode(2,OUTPUT);
    pinMode(3,OUTPUT);
    pinMode(4,OUTPUT);
    pinMode(5,OUTPUT);

    pinMode(6,INPUT);
    pinMode(7,INPUT);
    pinMode(8,INPUT);
    pinMode(9,INPUT);

    pinMode(13,OUTPUT);
}

void loop()
{
    digitalWrite(13,HIGH);
    delay(10);
    digitalWrite(13,LOW);

    W = digitalRead(6);
    X = digitalRead(7);
    Y = digitalRead(8);
    Z = digitalRead(9);
```

```
A = !W;  
  
B = 0;  
C = 0;  
D = 0;  
  
disp_dontcare(D,C,B,A);  
  
delay(1000);  
}
```

5 Output

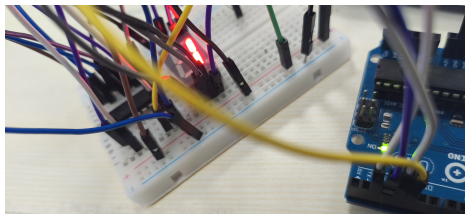


Figure 2: Output Showing 1

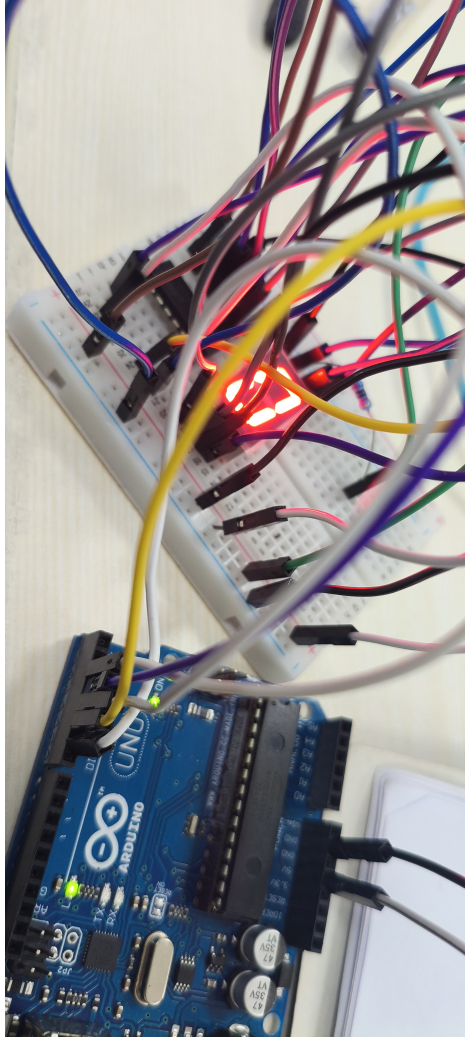


Figure 3: Output Showing 0

6 Result

The Boolean expression was implemented using Arduino Uno, IC 7474, IC 7447 and a seven segment display.

$$A = \bar{W}$$

For $W = 0$,

$$A = 1$$

Therefore the seven segment display shows 1.